Smart Robots Design

DEPARTMENT OF ENGINEERING
TOHOKU UNIVERSITY

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Potential Field and Mass Damper System on Mobile Robot (Hands on)

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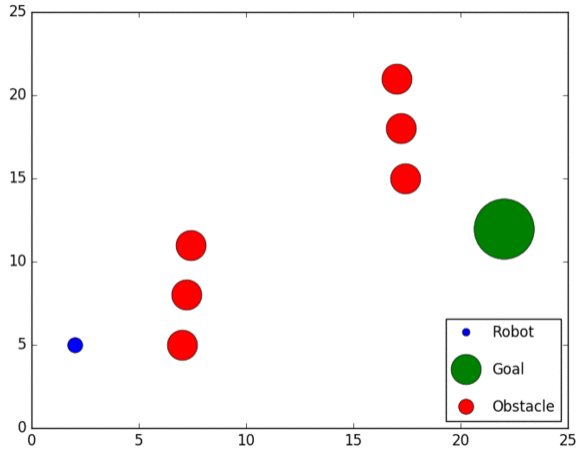
Problem

- How does a robot know when to stop?

Move toward goal

Avoid obstacles

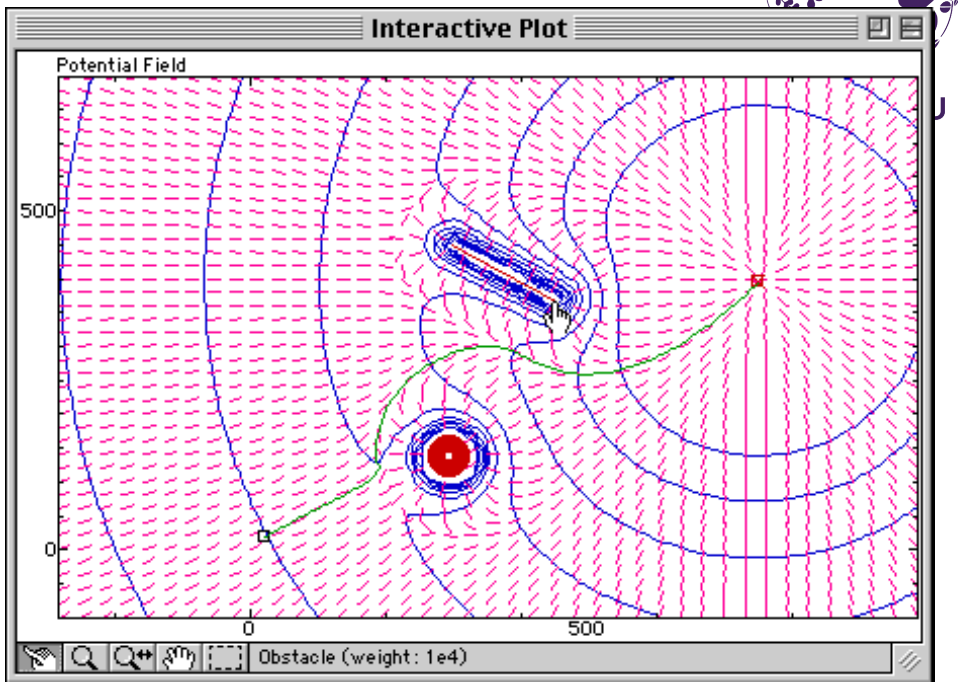
Do both at once

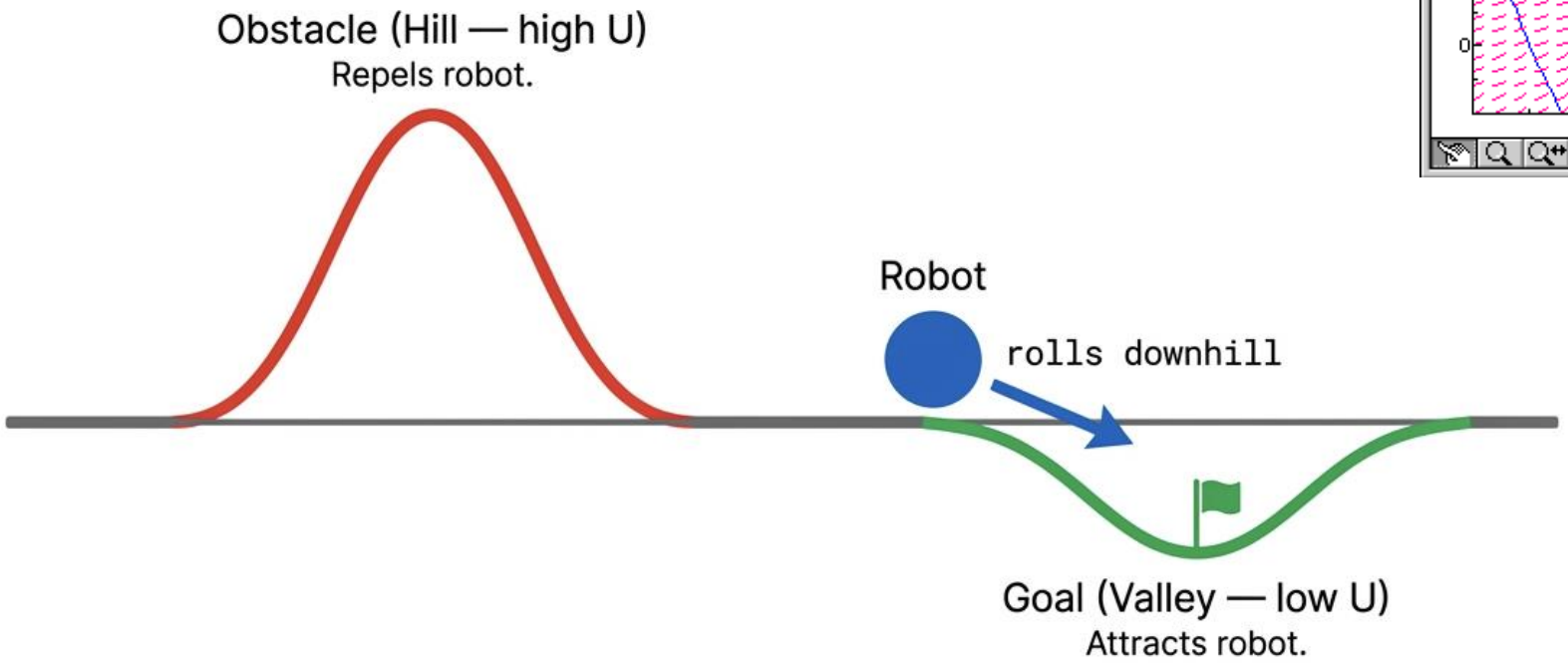


Potential Fields — a single mathematical framework that solves all three simultaneously

Concept

- Think of the robot as a ball on hilly landscape. It moves naturally to reduce the potential energy U(d).





Hill (obstacle)
High potential — repels robot

Valley (goal)
Low potential — attracts robot

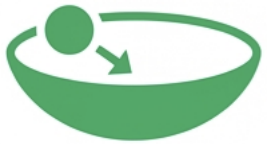
Ball
The robot

Rolling downhill
Always moves to reduce U(d)

Concept

Potential Energy U(d) where d = distance to obstacle

$$U_{total}(d) = U_{att}(d) + U_{rep}(d)$$



Attractive Potential U_att

$$U_{att} = \frac{1}{2} * k_{att} * (d - d_{goal})^2$$

Bowl shape — minimum exactly at d_goal.
The robot naturally rolls into this valley.

Repulsive Potential U_rep

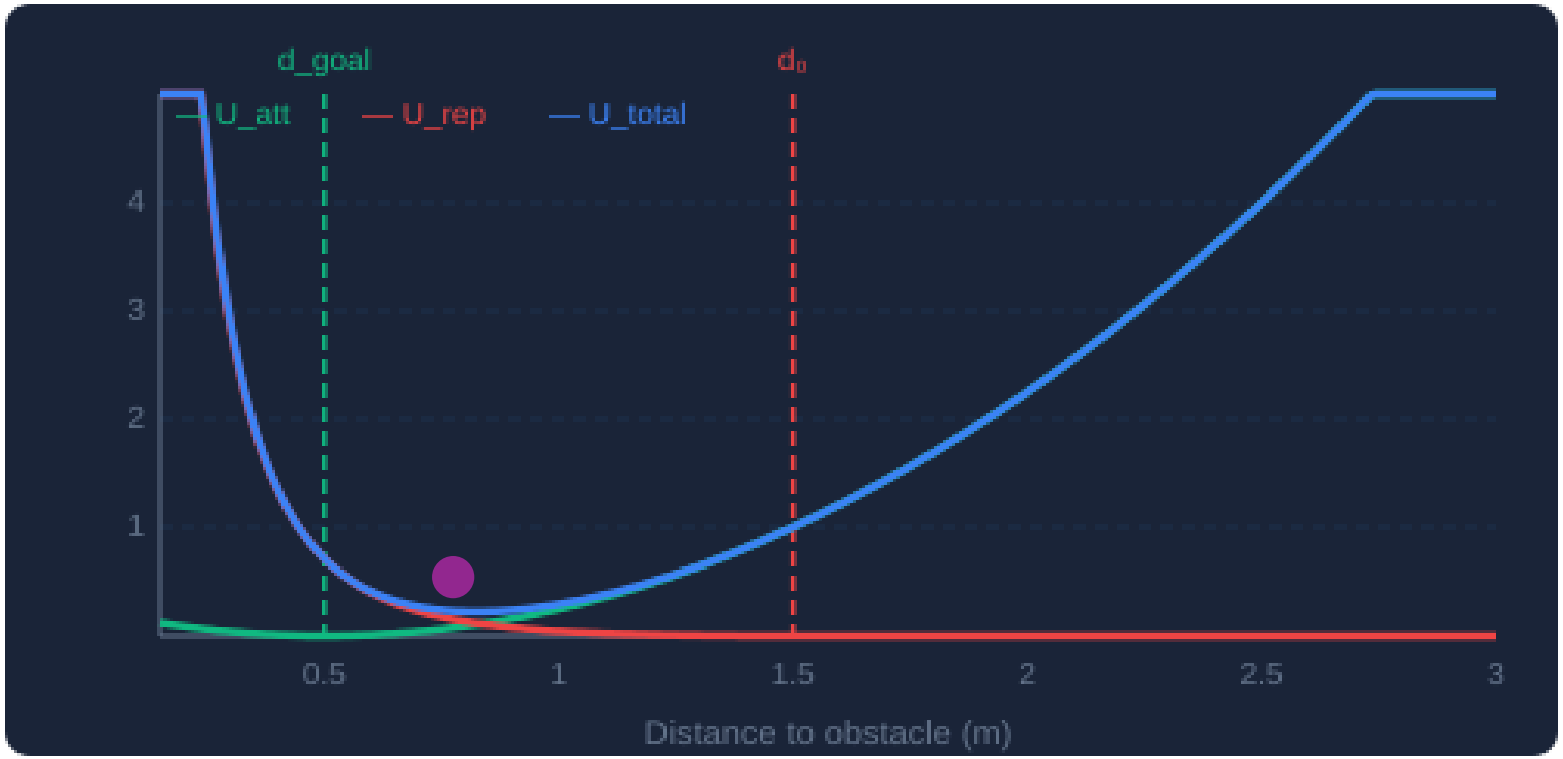
$$U_{rep} = \frac{1}{2} * k_{rep} * \left(\frac{1}{d} - \frac{1}{d_0}\right)^2$$

Steep hill — grows to infinity at d=0.
Active only when $d < d_0$ (influence radius).

The energy landscape U(d)

Potential Field lanscape

k_att=2.0 | k_rep=0.8 | d_0=1.5m | d_goal=0.5m



U_att is a parabola — bowl with minimum at d_goal

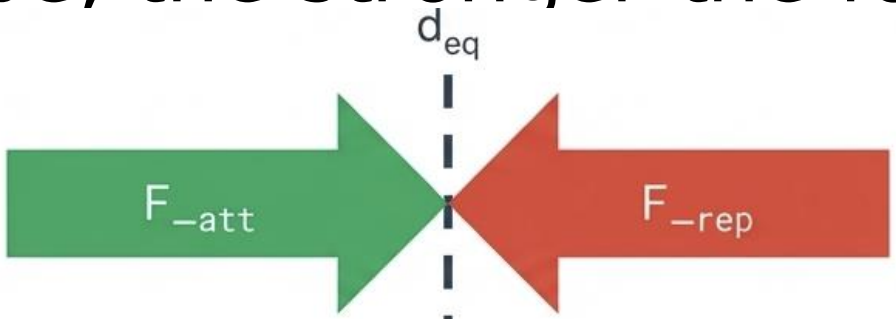
U_rep is zero outside d_0, grows steeply inside

U_total has a minimum — this is the equilibrium

Purple dot = where robot will stop

Forces and Equillibrium

- Force is the gradient of the landscape (F= -dU/dx). The steeper the slope, the stronger the force



Attractive Force

`F_att = k_att * (d - d_goal)`

Positive -> move forward. Negative -> move back. Zero at d_goal.

Repulsive Force

`F_rep = -k_rep * (1/d - 1/d_0) * (1/d^2)`

Always negative (pushes away). Infinity as d -> 0.

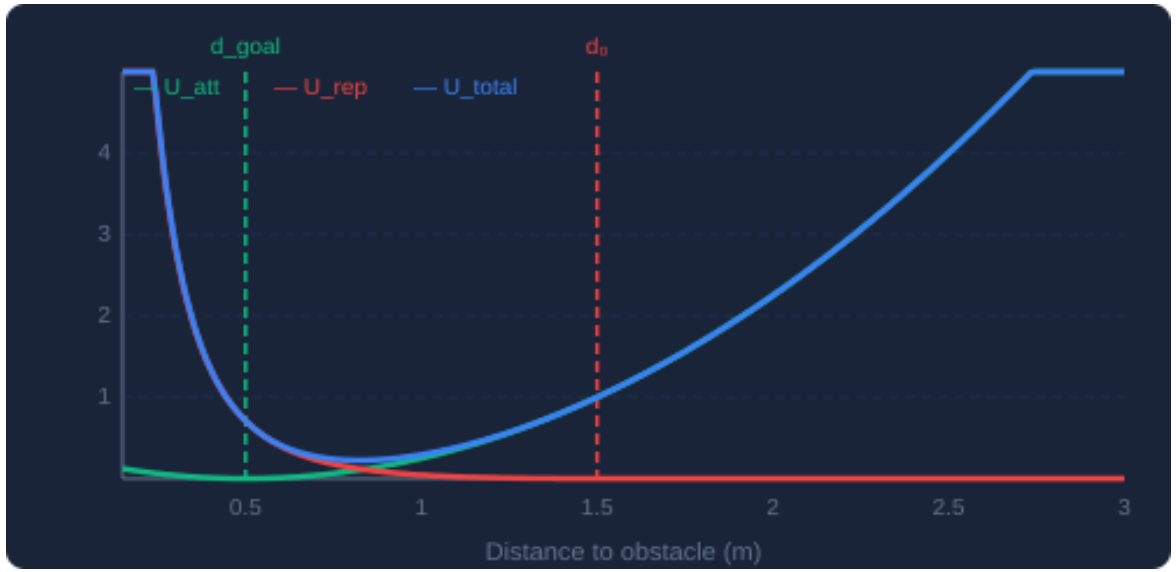
The Rule of Equilibrium: Robot stops when F_total = F_att + F_rep + F_damp = 0.

Key Insight: d_eq is NOT equal to d_goal. It depends entirely on the ratio of k_att to k_rep.

Equilibrium — where F_total = 0

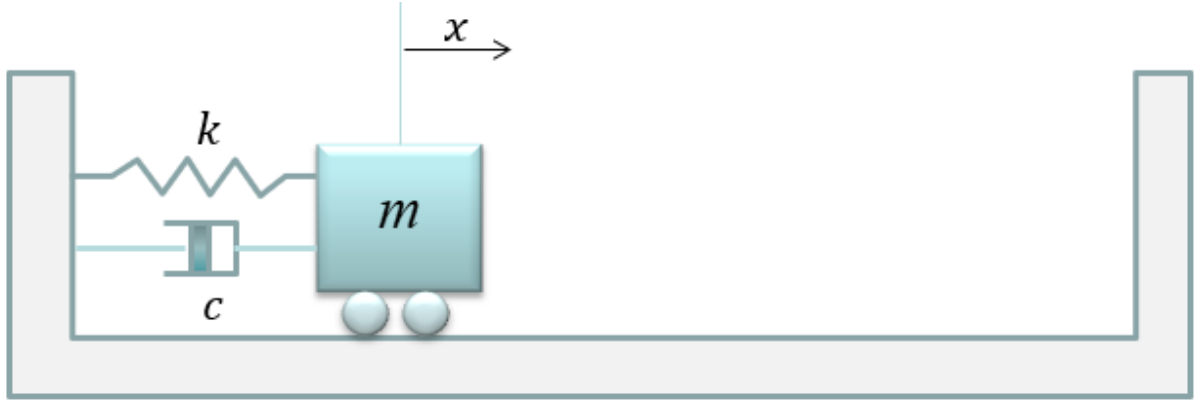
$$k_{att} \cdot (d_{eq} - d_{goal}) = k_{rep} \cdot (1/d_{eq} - 1/d_0) \cdot (1/d_{eq}^2)$$

01	d_eq is NOT equal to d_goal — it depends on the ratio k_att / k_rep
02	Increasing k_rep pushes d_eq further from the obstacle
03	Increasing k_att pulls d_eq closer to d_goal

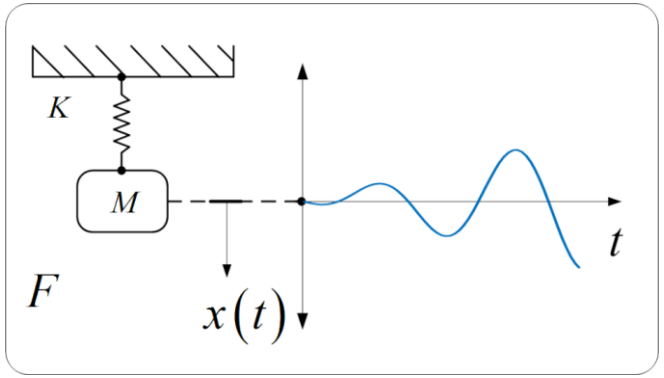


Spring mass damper

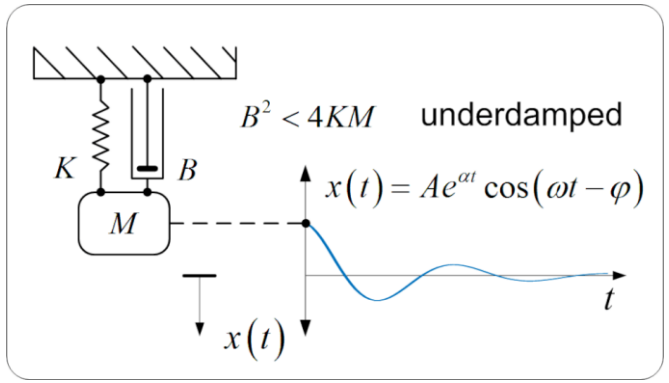
Same mathematics — same physics



$$m \cdot \ddot{x} + c \cdot \dot{x} + k \cdot x = 0$$



$$T = 2\pi / \sqrt{(k_{(att)} / m_{robot})}$$



$$F_s = -k(x - x_g)$$

$$F_d = -c\dot{x}$$

For the robot near an obstacle — identical equation:

$$m \cdot \ddot{x} + k_damp \cdot \dot{x} + k_att \cdot (d - d_goal) + F_rep = 0$$

Spring-Damper ↔ Robot System

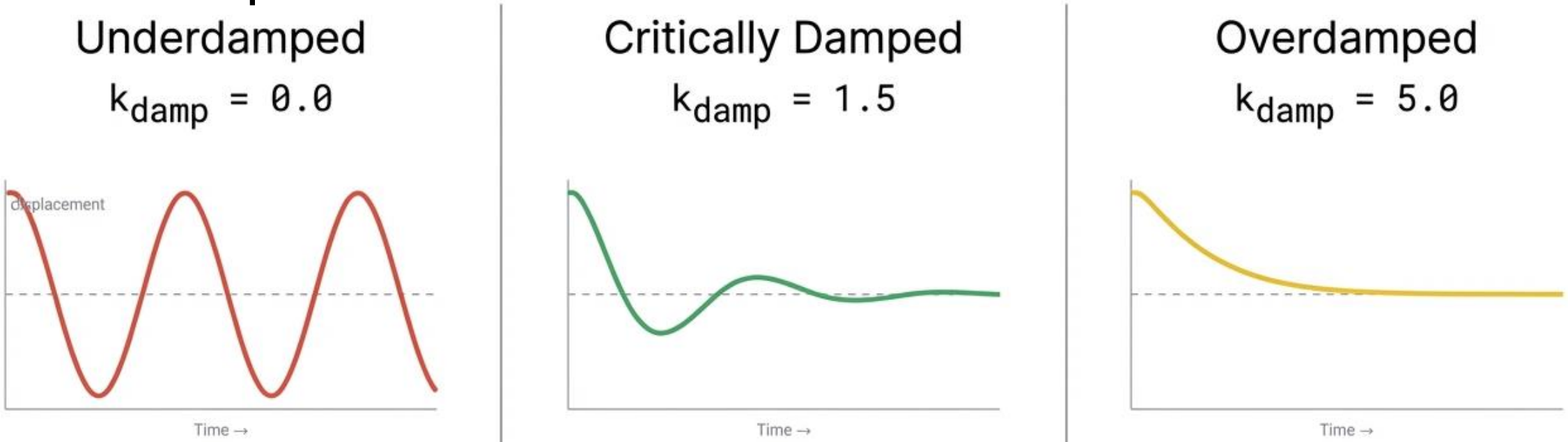
THE MAPPING

Spring-Mass System	Robot Parameter	Physical Meaning
Spring constant k	k_att	Strength of goal attraction
Rest length x_0	d_goal	Target stop distance
Mass m	m_robot ≈ 3.5 kg	Robot inertia
Damping coeff c	k_damp	Energy dissipation
Displacement	d − d_goal	Error from target distance
Critical damping $2\sqrt{(k \cdot m)}$	$2\sqrt{(k_att \cdot m)}$	Optimal k_damp ≈ 5.3

Key insight: Tuning k_damp on the robot is the same as adjusting damping on a physical spring system

Damping regimes

- Three regimes: Underdamped, Critically damped, Overdamped



Robot oscillates forever.
Never settles.

Smooth stop at equilibrium.
No overshoot — ideal!

Very slow approach.
Creeps to equilibrium.

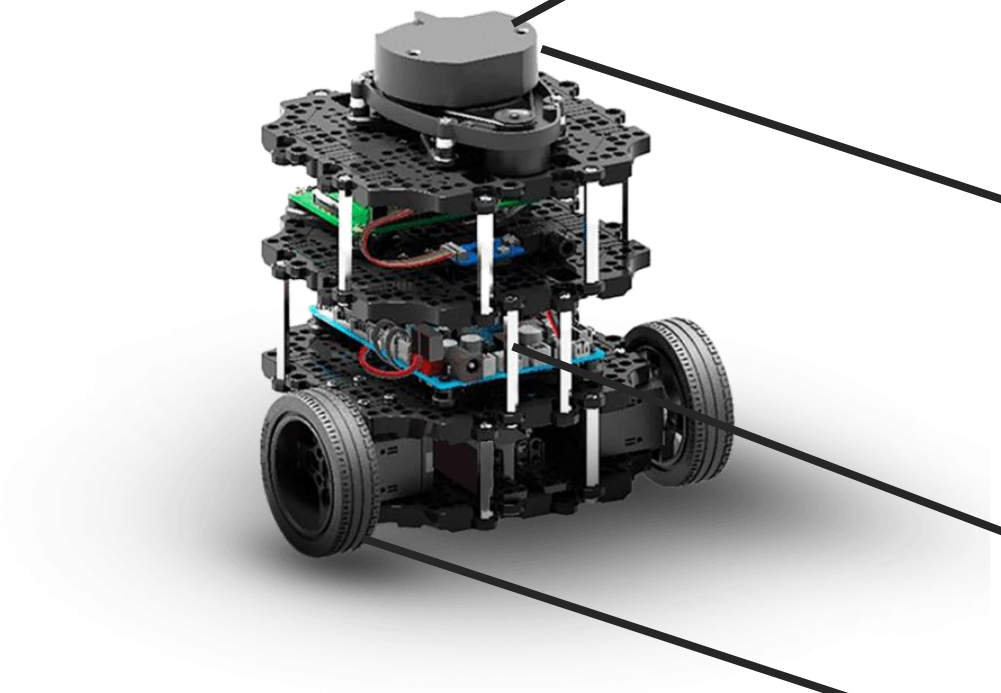
What controls what?

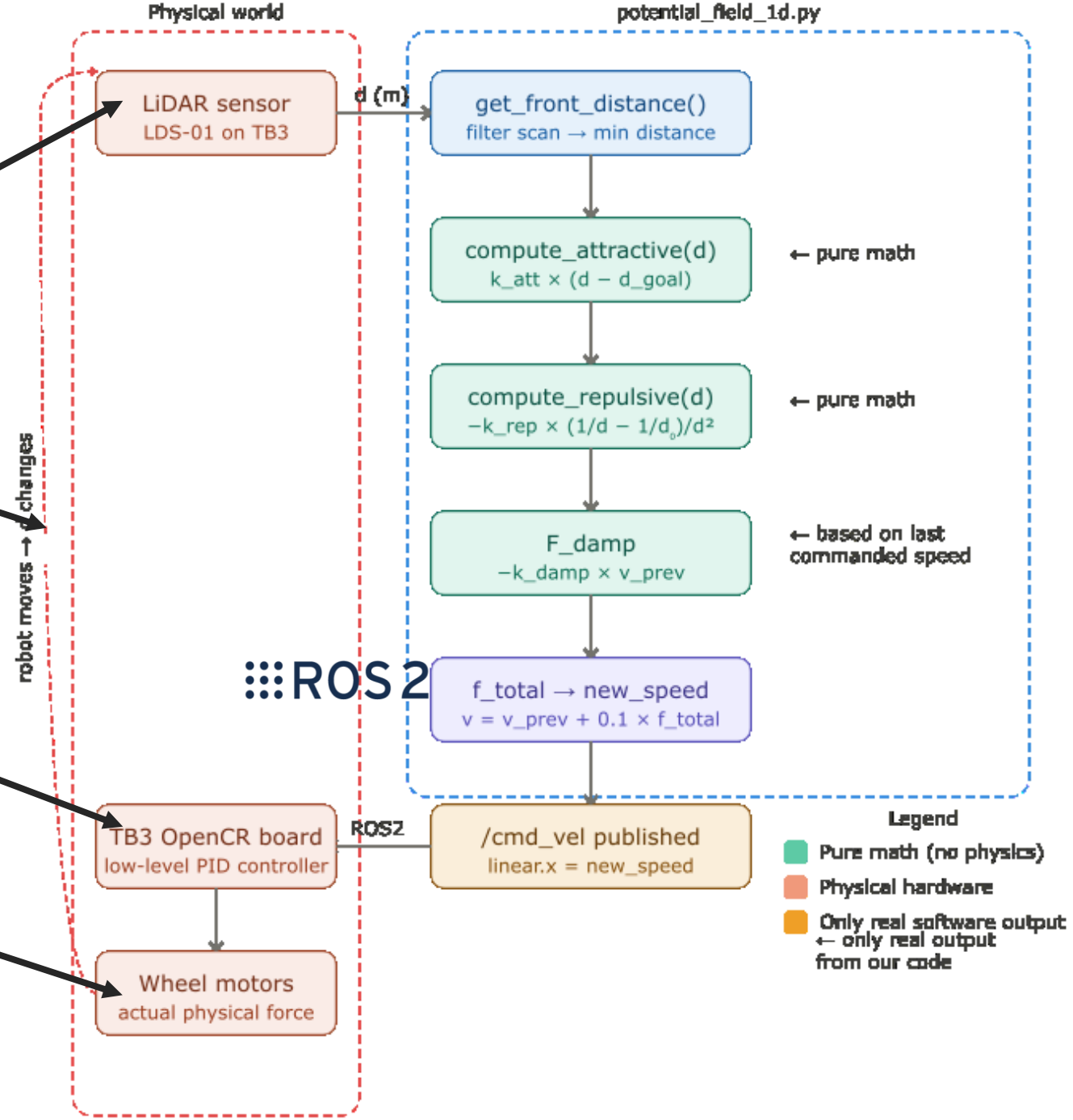
PARAMETER GUIDE

Change one at a time — predict before you run!

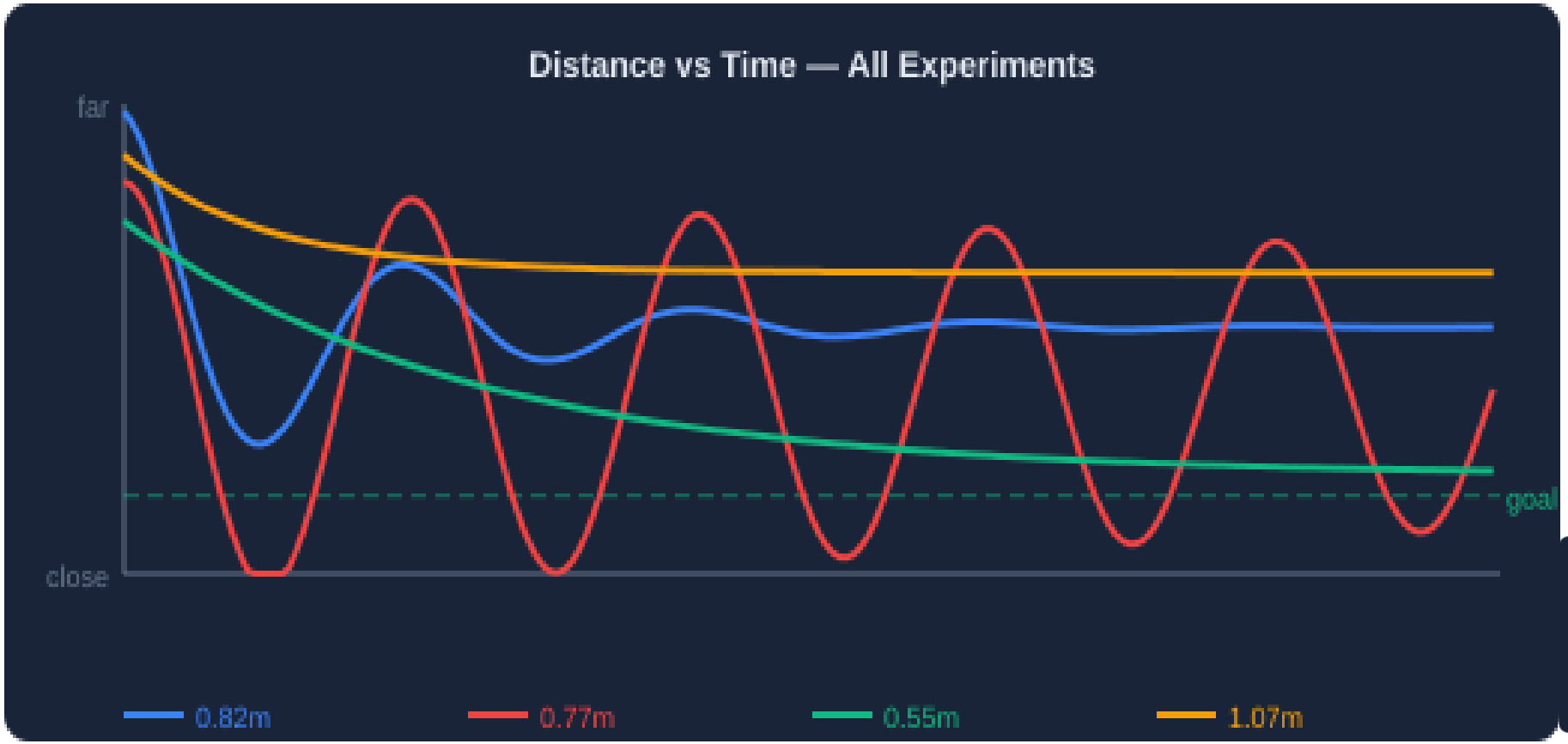
Parameter	Too small →	Too large →	Baseline
`k_att` Attractive gain	Weak pull — slow approach	Strong pull — overshoots	2.0
`k_rep` Repulsive gain	Gets too close — dangerous!	Stops far away from box	0.8
d_0 Influence radius	Late braking — may crash	Slows very early	1.5 m
`d_goal` Target distance	Equilibrium very close	Equilibrium far from box	0.5 m
`k_damp` Damping coefficient	Oscillates (underdamped)	Creeps slowly (overdamped)	1.5

What is happening



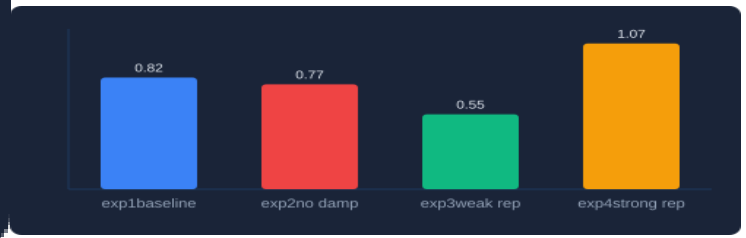


What should your results look like

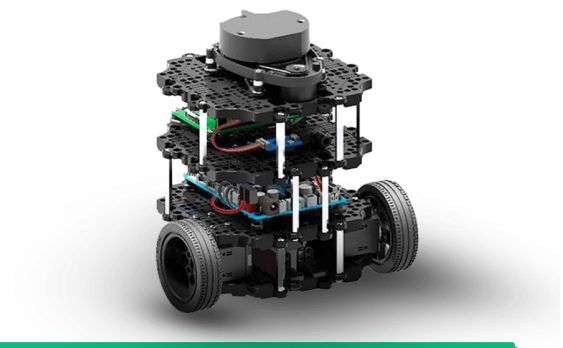


Measured equilibrium distances:

XXm	**Baseline**	Smooth stop ✓
XXm	**No damping**	Oscillated forever
XXm	**Weak repulsion**	Stopped close
XXm	**Strong repulsion**	Barely moved



Your Mission Today



20 min

Theory

Equations, analogy, parameters

20 min

Setup

Clone repo, build, connect robot

60 min

Experiments

Run, measure, predict, verify

Bonus- Final Challenge: Design your own parameters to stop exactly at 0.4m from the box — calculate, predict, verify!